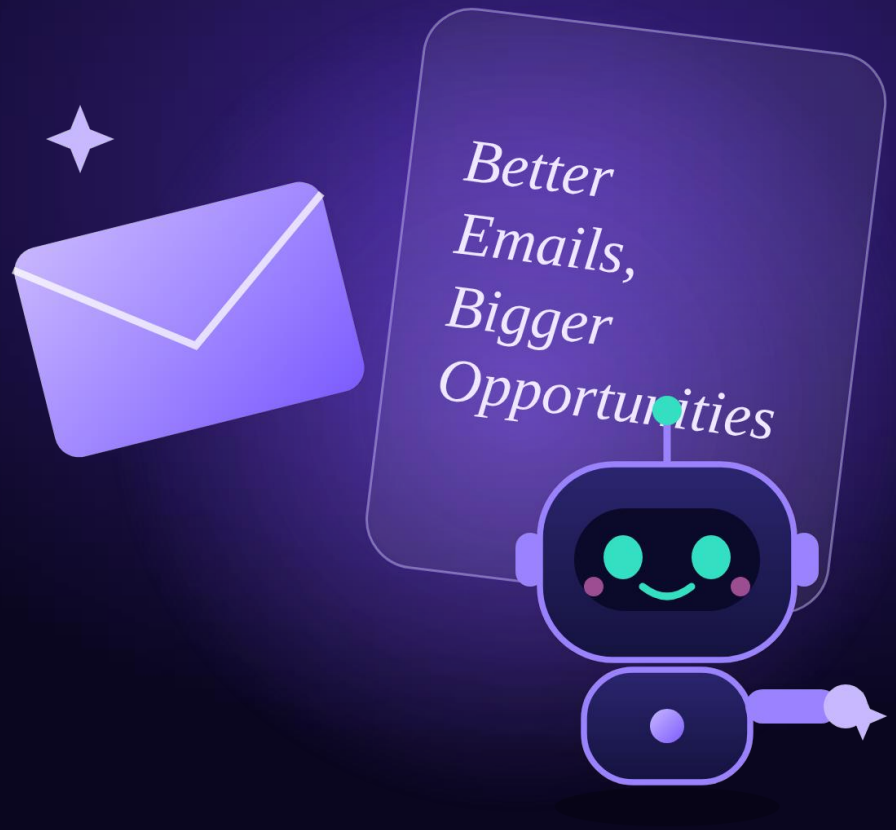
AI EMAIL GENERATOR

LetterLy

Turn your thoughts into perfect emails

FastAPI | Groq | Firebase | Vercel



Prepared by Anushay Ayat
SMIT (Saylani Mass IT Training), AI and Data Science
October 2026

What we will cover

01 **Objective and scope**
What the assignment asks and how LetterLy answers it

02 **How it works**
Request flow, tech stack and architecture

03 **Features**
Generation, AI tools and productivity features

04 **AI integration**
Prompt design and safe handling of output

05 **Design**
Brand, interface and responsive layout

06 **Data and privacy**
Storage, visitor spaces and secrets

07 **Quality**
Testing approach and results

08 **Delivery**
Deployment, challenges and next steps

The assignment and our answer

THE ASSIGNMENT ASKS FOR	LETTERLY DELIVERS
A FastAPI backend	**FastAPI app with 22 API endpoints**
Recipient name, purpose and tone	**Plus length, language and sender name**
An LLM writes the full email	**Groq (openai/gpt-oss-20b) in JSON mode**
Show it on an HTML page	**Landing page and a six-view dashboard**
GitHub link and a screenshot	**README, setup guide and deploy guide**

LetterLy at a glance

22

API endpoints

Generation, AI tools, data

10

Built-in templates

Plus your own saved ones

3

Tones

Professional, Friendly, Formal

7

Screens

Landing page and 6 dashboard views

3,200+

Lines of code

Backend, frontend and tools



English and Urdu

Right-to-left text in the page and in Word



Five ways to take it away

Copy, text, Word, PDF or your mail app



Voice and dark mode

Speak the purpose, switch themes any time

From idea to email in six steps

1

Fill the form

Recipient, purpose, tone, length, language

2

Send the request

Browser checks input, then calls the API

3

Build the prompt

FastAPI validates and writes the prompt

4

Ask the AI

Groq replies with a subject and a body

5

Parse and save

The JSON is checked and stored

6

Show the email

Typing effect, then edit or refine

While the AI works, the form locks and a progress label shows. When the email arrives it types out, and every field stays editable.

The technology behind it

FastAPI

Python web framework with 22 routes and input validation

Groq API

Fast LLM inference with openai/gpt-oss-20b

SQLite

Zero-setup database for local development

Firestore

Firebase cloud database used in production

python-docx

Builds Word files, including right-to-left Urdu

HTML, CSS, JS

No build step, fast pages, no framework needed

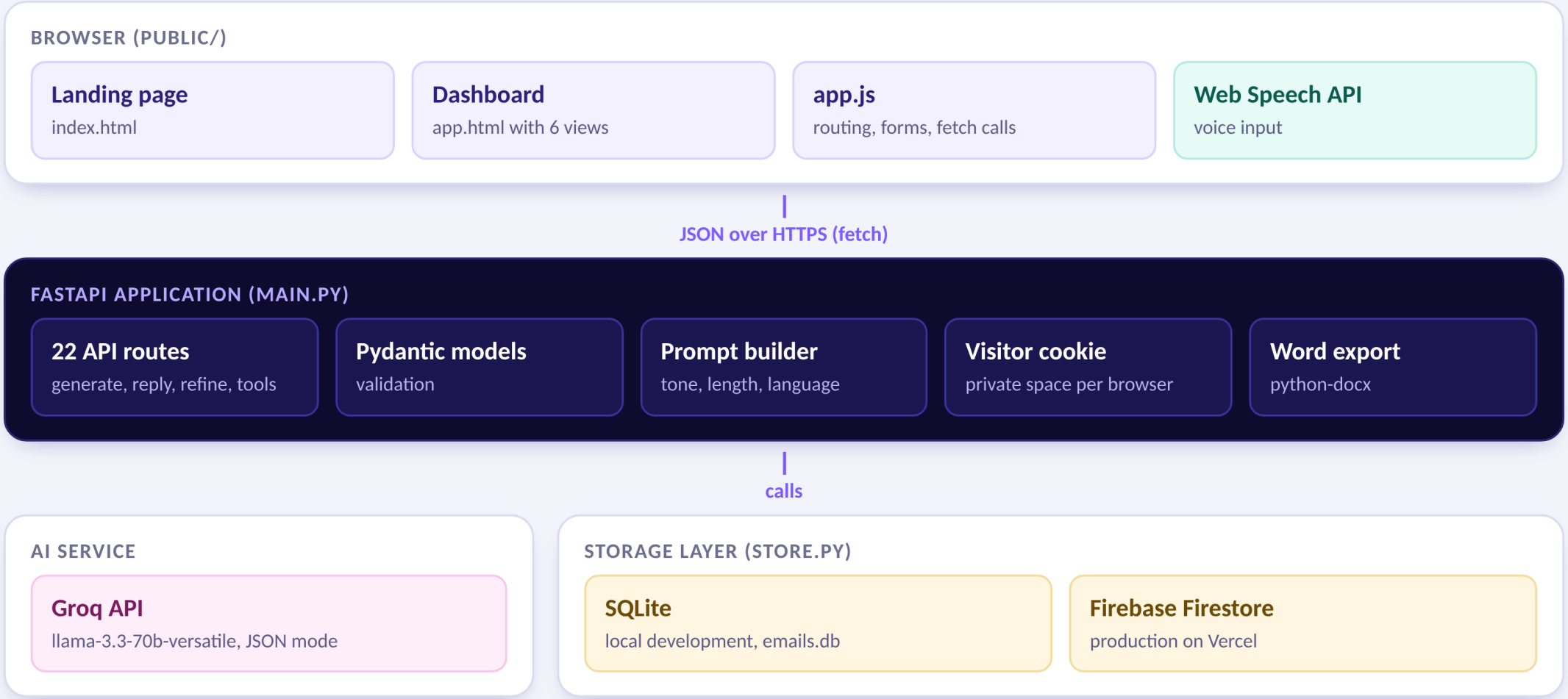
Web Speech API

Voice input in the browser, English and Urdu

Vercel

Hosting with a CDN for pages and a function for the API

System architecture



Generate an email

Recipient name

Who the email is for

Email purpose

Type it, speak it or start from a template

Tone

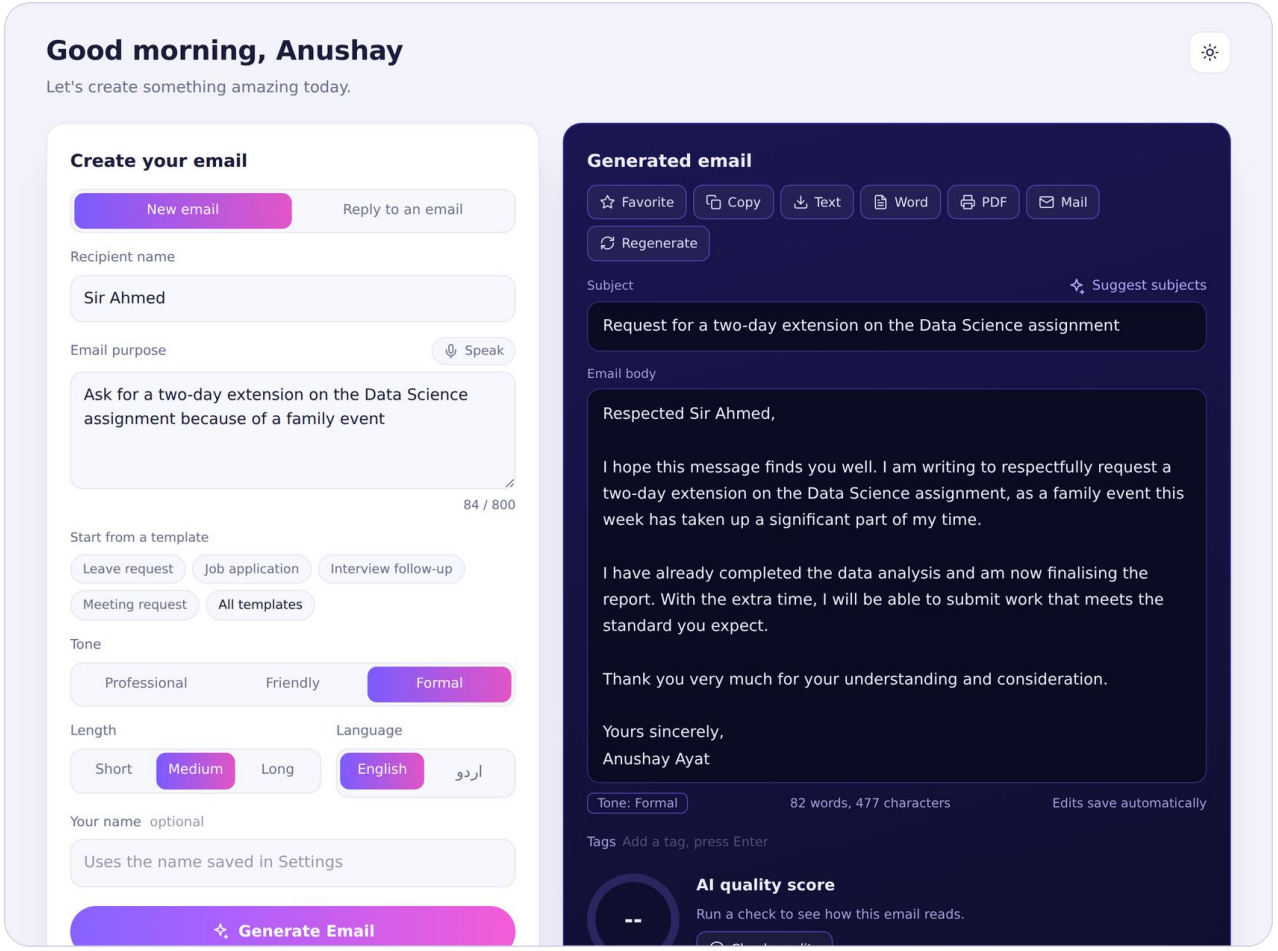
Professional, Friendly or Formal

Length

Short, Medium or Long

Language

English or Urdu



The dashboard with sample data: form on the left, generated email on the right.

Eight AI tools around every draft

Reply mode

Paste a received email and get a reply

Custom changes

Type "make it more polite" and it rewrites

Subject ideas

Three alternative subject lines to pick from

Quality score

A 0 to 100 score with practical tips

Tone detector

Percentages across four tones

Grammar fix

Corrects spelling and punctuation only

Thread summary

Gist, questions and deadlines in a long thread

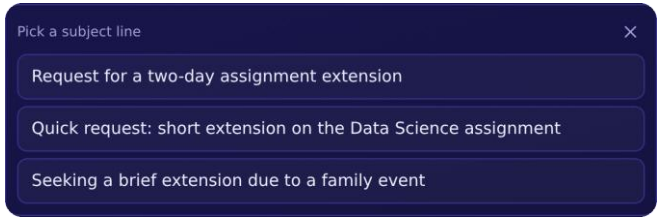
Quick refine

Shorter, longer or translate to Urdu

The AI tools in action

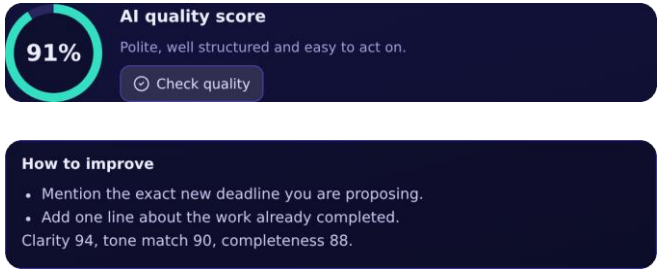
Subject suggestions

Three options, one click to apply



Quality score and tips

A score, a verdict and what to improve



Tone detector

How formal, friendly or confident it sounds



Each tool is a single API call: /subjects, /analyze and /tone. The quality score is saved with the email, so it is still there when you reopen it from history.

Prompt design and AI integration

1. **System prompt**
   Sets the role and demands a JSON object

2. **User prompt**
   Recipient, purpose, tone guide, length guide

3. **JSON mode**
   Groq is asked for a JSON response

4. **Safe parsing**
   Tolerant parser, clear error if unreadable

Task	Temperature	Why
Generate email	0.7	Natural, varied wording
Refine or translate	0.5	Change it, keep the facts
Subject ideas	0.9	Three clearly different options
Quality score	0.3	Steady, honest scoring
Tone detector	0.2	Consistent measurement
Summary	0.2	Faithful to the thread
Grammar fix	0.1	Fix only, never rewrite

Pasted emails and threads are marked as content to respond to, so instructions hidden inside them are not followed.

Built to save you time

Templates
10 built in, plus templates you save

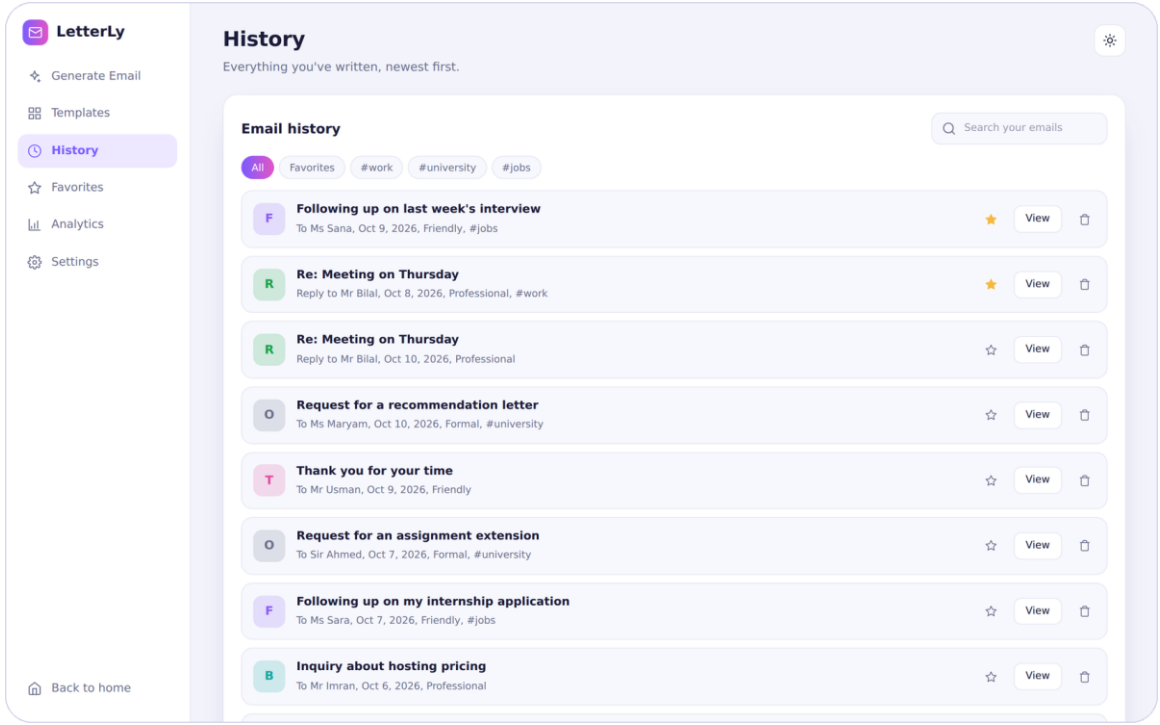
Favorites and tags
Star emails and add up to five tags

Searchable history
Search and filter by tag or favorite

Export
Copy, text, Word, PDF or mail app

Voice input
Speak the purpose in English or Urdu

Analytics
Emails, quality, tone mix, weekly chart



History view with sample data

10 templates

5 export options

2 voice languages

A design system with two themes

Violet	Pink	Teal	Ink	Night	Mist
#7C5CFF	#E255C7	#35E0C2	#1B1B3A	#0A0620	#F3F4FB

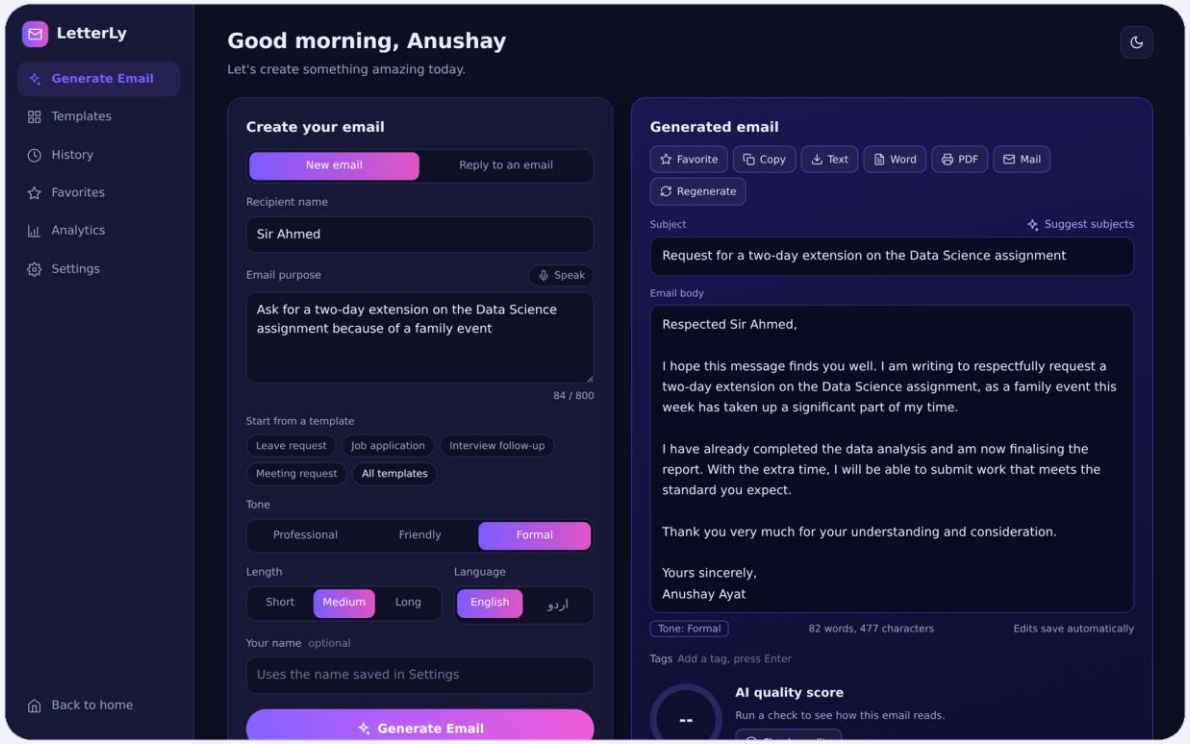
TYPOGRAPHY

Headings: Playfair Display (serif)

Body: Inter (clean sans)

PRINCIPLES

- Rounded cards and soft shadows
- One gradient accent, used sparingly
- Large touch targets and clear focus rings



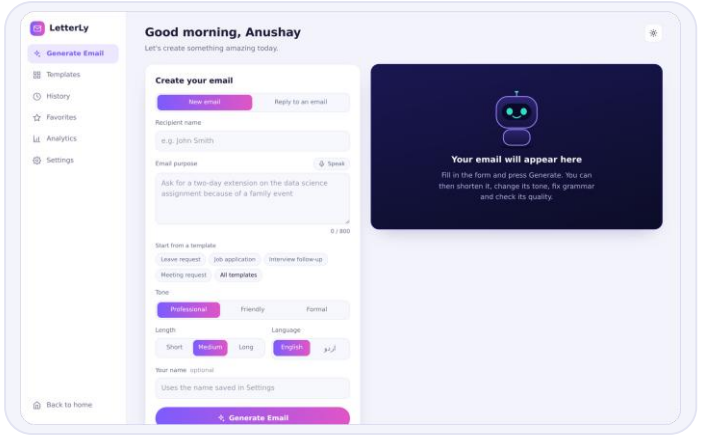
The same dashboard in the dark theme (sample data)

Keyboard focus, labelled controls and reduced-motion support are built in.

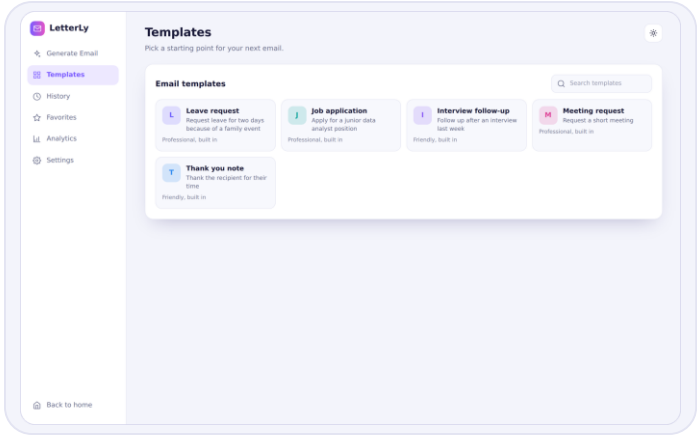
Screens across the product



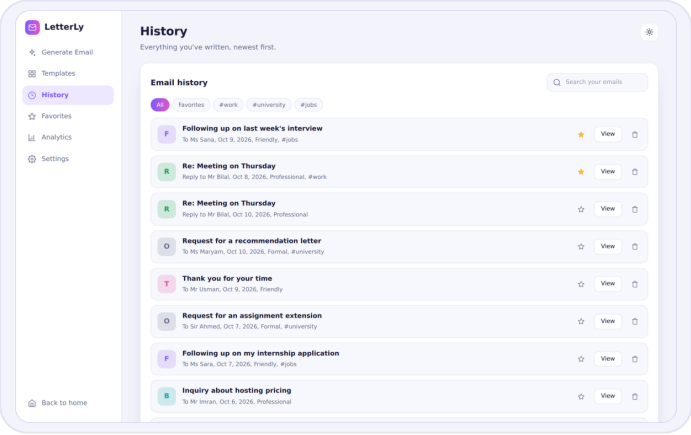
Landing page



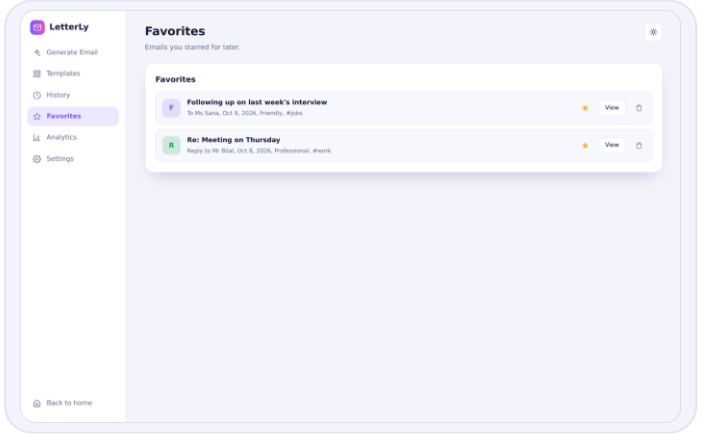
Dashboard



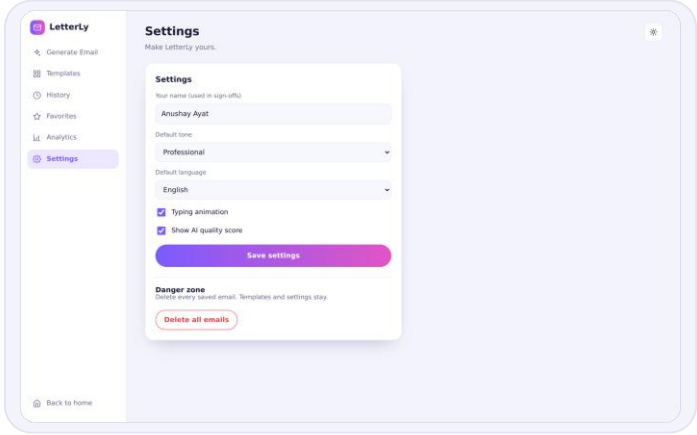
Templates



History

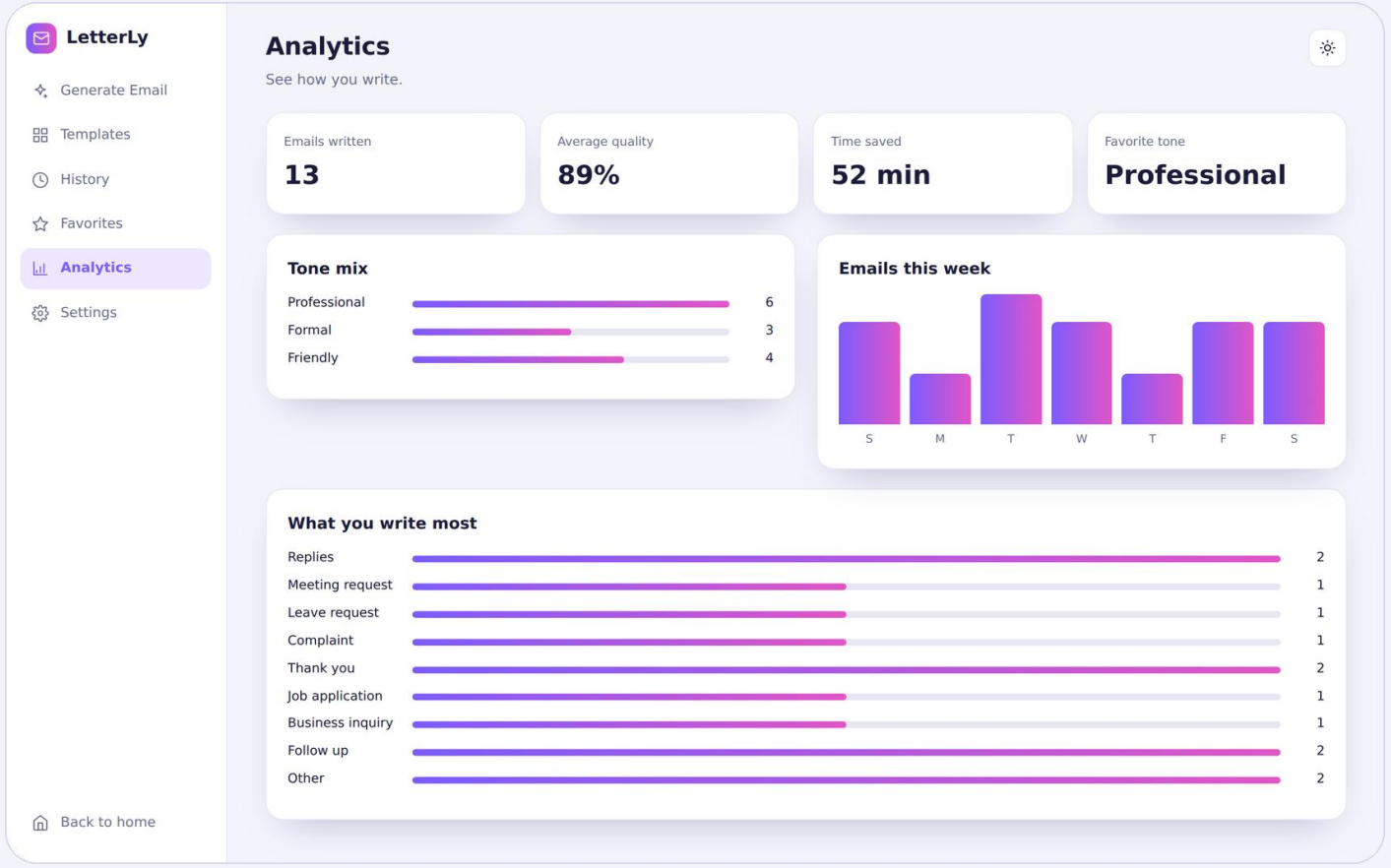


Favorites



Settings

Analytics and settings



Analytics view with sample data

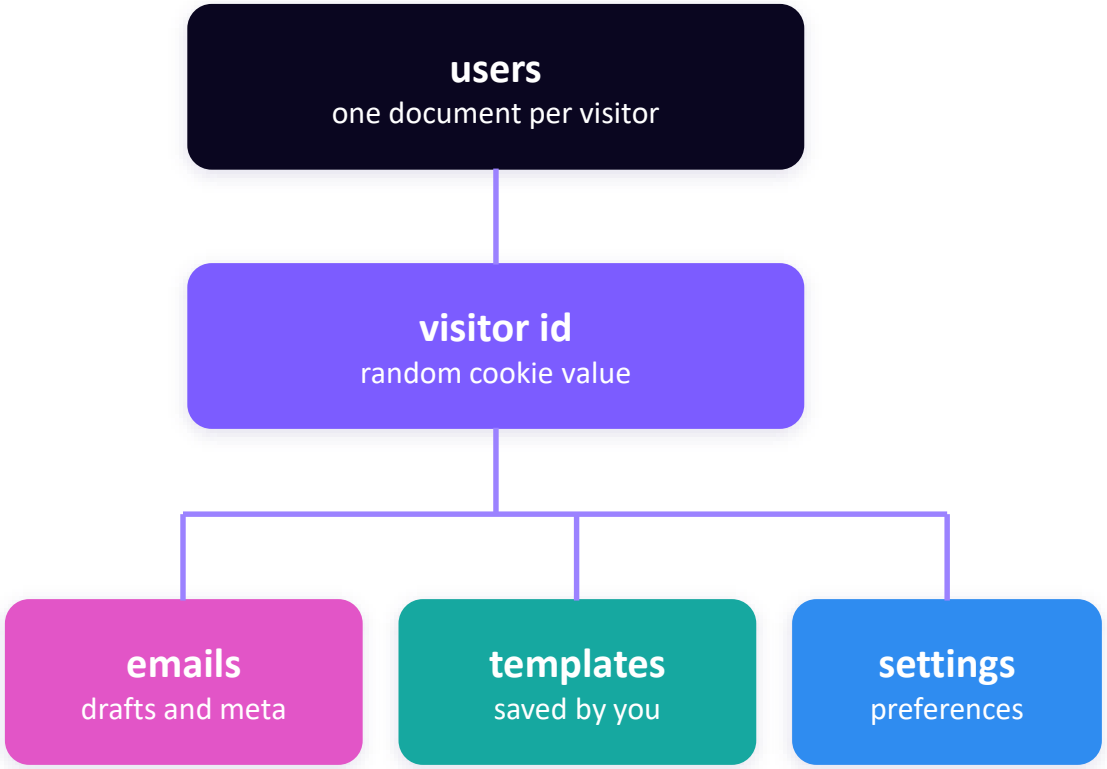
Analytics

- Emails written and time saved
- Average AI quality score
- Tone mix and what you write most
- Emails per day this week

Settings

- Name used in sign-offs
- Default tone and language
- Typing animation on or off
- Quality score on or off

Data, privacy and security



Firestore layout. Locally, SQLite keeps the same data in tables.

Secrets stay out of code
Keys live in environment variables

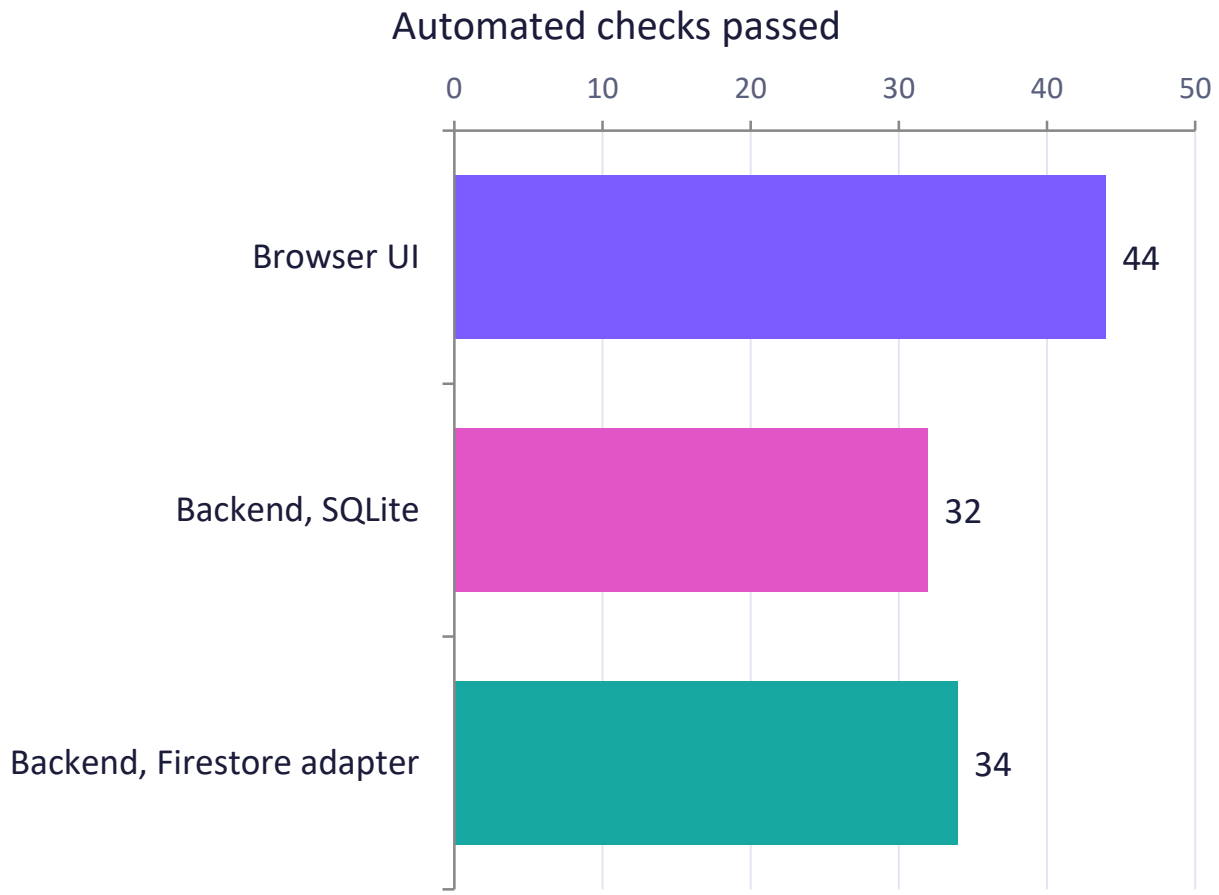
Keys never reach GitHub
.gitignore blocks .env and key files

Private space per visitor
Cookie is HttpOnly, Secure, SameSite=Lax

Server-only database access
Firestore rules stay locked

Input limits everywhere
Pydantic checks every request

Testing and quality

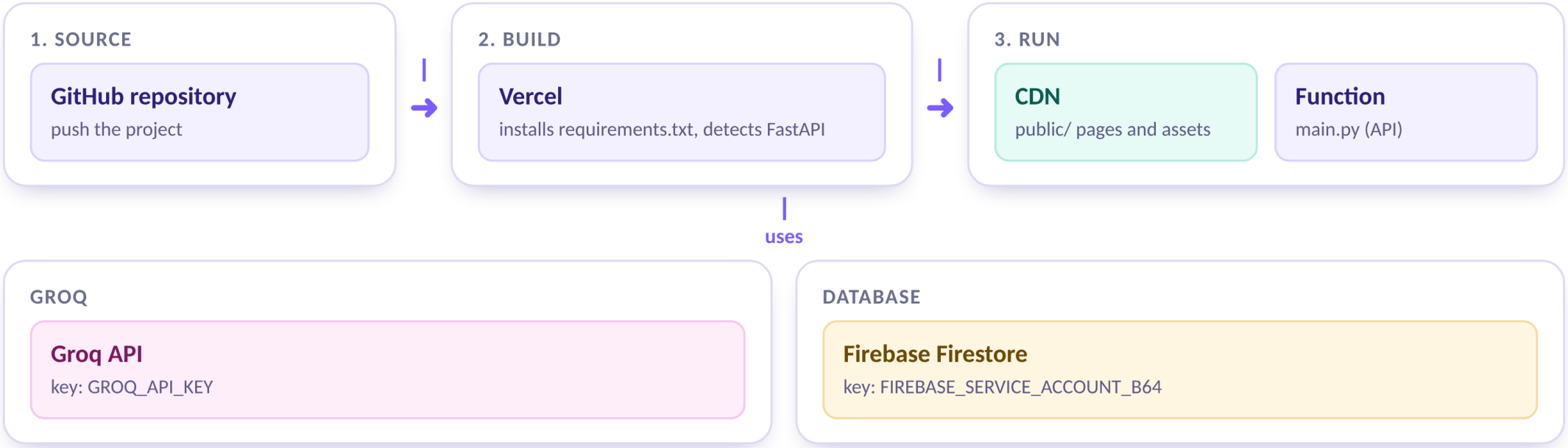


110 **checks, all passing**
across three test runs

What is covered

- Every dashboard view and AI tool
- Reply mode, templates, export hooks
- Private visitor spaces and clearing data
- Phone layout without sideways scroll

Deploying on Vercel with Firebase



Add GROQ_API_KEY

Your Groq key, kept in the Vercel settings

Add the Firebase key

FIREBASE_SERVICE_ACCOUNT_B64, made with the encoder tool

Check /api/health

It should report storage: firestore

Challenges and how we solved them

Challenge	Solution
Vercel's disk is read-only	A storage layer picks Firestore in production and keeps SQLite for local work
The AI does not always return clean JSON	JSON mode, a tolerant parser and a clear error message
Pasted emails could contain instructions	Prompts mark them as content to answer, not commands
A public link would share one database	A cookie gives every browser a private space
Urdu in Word and PDF	Right-to-left support in the page, in Word, and a print-only sheet for PDF
The page must feel alive while the AI works	Locked controls, progress labels and a typing effect

Where LetterLy can go next

Next

Sign-in

Accounts instead of cookie spaces

Rate limits

Protect the shared AI key

Streaming

Show the email as it is written

Later

More languages

Arabic, Hindi and more

Send from Gmail

Mail provider integration

Team templates

Share templates with a group

Ideas

Subject A/B tests

Compare open rates

Calendar-aware replies

Suggest meeting times

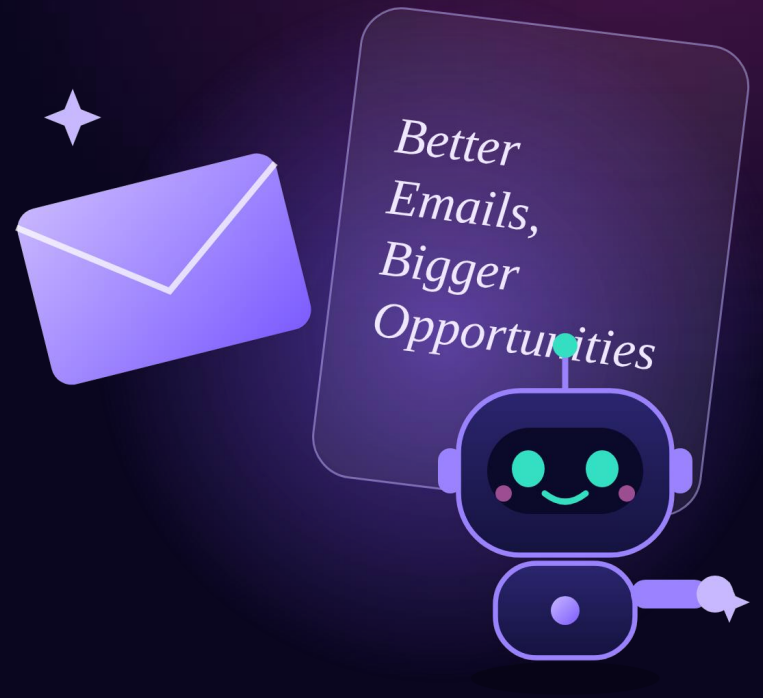
Installable app

A phone-friendly web app

Thank you

Questions?

- A complete product, not just a form: 22 endpoints and 7 screens
- AI used with care: structured prompts, validation and safe handling
- Ready to ship: tested, documented and set up for Vercel and Firebase



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